

Advome Deployment Guide

Overview

This guide covers deploying the Advome platform from local development to production.

Architecture

```
Frontend (GitHub Pages / Netlify / Vercel)
  ↓ HTTPS
Backend API (Heroku / Render / AWS / DigitalOcean)
  ↓
Database (PostgreSQL — Supabase / AWS RDS / Heroku Postgres)
  ↓
Document Storage (AWS S3)
  ↓
AI Engine (Kimi API — Moonshot AI)
  ↓
Payment Gateway (PayFast)
```

Prerequisites

- Node.js 18+ installed locally
- PostgreSQL 14+ installed locally (for development)
- AWS account (for S3 document storage)
- Kimi API key from Moonshot AI
- PayFast merchant account (for South Africa)
- GitHub account (for frontend hosting)

Step 1: Local Development Setup

1.1 Clone Repository

```
git clone https://github.com/KgothatsoMmoledi/advome.git
cd advome
```

1.2 Install Backend Dependencies

```
mkdir backend
cd backend
cp ../server.js .
cp ../package.json .
cp ../.env.example .env
npm install
nano .env # Edit with your values
```

1.3 Set Up Database

```
createdb advome_db
npm run migrate
npm run seed # Optional
```

1.4 Start Backend Server

```
npm run dev
# Server runs on http://localhost:3000
```

1.5 Test Frontend Locally

```
cd ..
python3 -m http.server 8080
# Frontend runs on http://localhost:8080
```

Step 2: Frontend Deployment (GitHub Pages)

2.1 Enable GitHub Pages

1. Go to repository Settings → Pages
2. Source: Deploy from a branch
3. Branch: main / root
4. Click Save
5. Site will be at: <https://kgothatsommoledi.github.io/advome/>

2.2 Update API URLs in Frontend

Edit all HTML files to point to production API:

```
const API_BASE_URL = 'https://api.advome.co.za/v1'; // Production
```

Step 3: Backend Deployment (Render.com)

3.1 Create Render Account

1. Go to <https://render.com>
2. Sign up with GitHub
3. Create New Web Service

3.2 Deploy Backend

1. Connect your GitHub repository
2. Select the backend directory
3. Build Command: npm install
4. Start Command: node server.js
5. Add environment variables from .env
6. Click Create Web Service

3.3 Create PostgreSQL Database

1. In Render Dashboard, click New → PostgreSQL
2. Name: advome-db
3. Plan: Free (or Starter for production)
4. Copy Internal Database URL to env vars

Step 4: AWS S3 Setup

4.1 Create S3 Bucket

1. AWS Console → S3
2. Create bucket: advome-documents
3. Region: af-south-1 (Cape Town)
4. Block all public access: Yes
5. Enable versioning: Yes

4.2 Create IAM User

1. IAM → Users → Create user
2. Name: advome-api-user
3. Attach policy: AmazonS3FullAccess
4. Save Access Key ID and Secret Access Key
5. Add to .env

Step 5: Kimi API Setup

5.1 Get API Key

1. Go to <https://platform.moonshot.cn>
2. Create account and verify
3. Go to API Keys section
4. Generate new key
5. Add to .env as KIMI_API_KEY

5.2 Test Connection

```
curl -X POST https://api.moonshot.cn/v1/chat/completions -H "Authorization: Bearer YOUR_KIMI_API_KEY"
↪ -H "Content-Type: application/json" -d '{"model": "kimi-latest", "messages": [{"role": "user",
↪ "content": "Hello"}]}'
```

Step 6: PayFast Integration

6.1 Create PayFast Account

1. Go to <https://www.payfast.co.za>
2. Register as a merchant
3. Complete verification (FICA documents)
4. Get Merchant ID and Merchant Key

6.2 Configure PayFast

Add to .env:

```
PAYFAST_MERCHANT_ID=10000100
PAYFAST_MERCHANT_KEY=your_key
PAYFAST_PASSPHRASE=your_passphrase
PAYFAST_SANDBOX=true
```

Step 7: Domain & SSL

7.1 Register Domain

Register advome.co.za at <https://co.za> or <https://domains.co.za>

7.2 Configure DNS

1. Point A record to backend server IP
2. Point CNAME (www) to frontend hosting

7.3 SSL Certificate

Use Let's Encrypt (free) via Certbot or Cloudflare (free SSL + CDN)

Step 8: Production Checklist

Security

- JWT secret is strong (min 32 chars)
- API keys in environment variables only
- Database credentials not in code
- CORS restricted to production domains
- Rate limiting enabled
- File upload size limits set
- Input validation on all endpoints
- SQL injection prevention
- XSS protection (Helmet.js)

POPIA Compliance

- Privacy Policy published and linked
- Terms & Conditions published and linked
- Data encryption at rest and in transit
- User consent mechanism
- Data deletion capability
- Breach notification procedure
- Information Officer appointed

Performance

- Database indexes on frequent queries
- S3 CDN for document delivery
- API response caching
- Frontend asset minification
- Image optimization

Monitoring

- Error tracking (Sentry)
- Uptime monitoring (UptimeRobot)
- API analytics (PostHog)

- Server logs (CloudWatch)

Cost Estimate (Monthly)

Service	Development	Production Start	Production Scale
Frontend	Free	Free	R500
Backend	Free	R500	R2,000
Database	Free	Free	R1,500
Storage	Free	R200	R1,000
Kimi API	Free tier	R500	R2,000
Domain	—	R150	R150
Email	Free	R200	R500
Monitoring	Free	R300	R1,000
Total	R0	R1,850	R8,650

Revenue at 100 users: R3,000/month Revenue at 500 users: R15,000/month Revenue at 50 attorneys: R2,500/month

Next Steps After Deployment

1. Register company: Advome (Pty) Ltd at CIPC
2. Open business bank account
3. Register for VAT if revenue exceeds R1M/year
4. Appoint Information Officer for POPIA
5. Get professional indemnity insurance
6. Consult attorney to review T&Cs annually
7. Build user community (WhatsApp/forum)

Support

For deployment issues: support@advome.co.za